

5.13.67

EE25BTECH11042 - Nipun Dasari

Question:

For what value of k do the following system of equations possess a non trivial (i.e., not all zero) solution over the set of rationals $\mathbb{Q}$?

$$x + ky + 3z = 0 \quad (0.1)$$

$$3x + ky - 2z = 0 \quad (0.2)$$

$$2x + 3y - 4z = 0 \quad (0.3)$$

$$(0.4)$$

Solution: Given:

$$\begin{pmatrix} 1 \\ k \\ 3 \end{pmatrix}^T \mathbf{x} = 0 \quad (0.5)$$

$$\begin{pmatrix} 3 \\ k \\ -2 \end{pmatrix}^T \mathbf{x} = 0 \quad (0.6)$$

$$\begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix}^T \mathbf{x} = 0 \quad (0.7)$$

Combining (0.5),(0.6),(0.7):

$$A\mathbf{x} = 0 \quad (0.8)$$

where A is :

$$\begin{pmatrix} 1 & k & 3 \\ 3 & k & -2 \\ 2 & 3 & -4 \end{pmatrix} \quad (0.9)$$

For this system to have non-trivial solution these planes must intersect on a line, i.e., any two of the rows must be linearly dependent in order to have infinitely many non-trivial solutions. We simplify the matrix by using gaussian elimination:

$$\begin{pmatrix} 1 & k & 3 \\ 3 & k & -2 \\ 2 & 3 & -4 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - 3R_1} \begin{pmatrix} 1 & k & 3 \\ 0 & -2k & -11 \\ 2 & 3 & -4 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - 2R_1} \begin{pmatrix} 1 & k & 3 \\ 0 & -2k & -11 \\ 0 & 3 - 2k & -10 \end{pmatrix} \quad (0.10)$$

Now, we continue the row reduction to get the matrix into row echelon form:

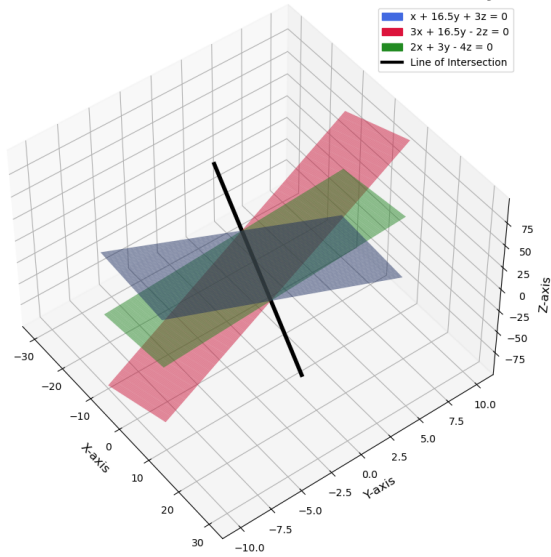
$$\begin{pmatrix} 1 & k & 3 \\ 0 & -2k & -11 \\ 0 & 3-2k & -10 \end{pmatrix} \xleftrightarrow{R_3 \leftarrow -2kR_3 + (3-2k)R_2} \begin{pmatrix} 1 & k & 3 \\ 0 & -2k & -11 \\ 0 & 0 & 2k-33 \end{pmatrix} \quad (0.11)$$

For a non-trivial solution, the third element of the last row must be zero.

$$2k - 33 = 0 \quad (0.12)$$

$$\Rightarrow k = \frac{33}{2} \quad (0.13)$$

Intersection of Three Planes at k = 16.5 (from C library)



Intersection of Three Planes when $k = 16.5$ (Line of Solutions)